

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

*I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

(5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

(5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

(5 Marks)

- (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
- (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
- (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
- (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

- i) Identify the epoch at which the model begins to overfit.
- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
- i) Identify the epoch at which the model starts overfitting.
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

- (a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
- (b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
- (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
- (d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
- (e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
- (f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
- (g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
- (h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
- i) At which epoch does the model achieve optimal generalization?
- ii) Identify signs of overfitting from the data.
- iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

- a) Front identified as Front: 119
- b) Front identified as Back: 06
- c) Back identified as Back: 120
- d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

- a) Diseased identified as Diseased: 138
- b) Diseased identified as Healthy: 12
- c) Healthy identified as Healthy: 135
- d) Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective):
(5 Marks)

- Defective identified as Defective: 230
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.
(5 Marks)

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.
(5 Marks)

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing.
(5 Marks)

Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity.
(5 Marks)

15. A deep learning model is trained for a classification task, and the following loss values are observed:
(5 Marks)

- Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
- Epoch 2: Training Loss = 2.4, Validation Loss = 2.2
- Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
- Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
- Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
- Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
- Epoch 7: Training Loss = 1.3, Validation Loss = 2.4

viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

- At which epoch does the model achieve the best generalization performance?
- At which point does overfitting begin? Justify your answer.
- Suggest two techniques to reduce overfitting and briefly explain how they help.

ANSWERS:

1. Bayes Theorem (Box C)

Given:

- $P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$
- $P(\text{Blue} | A) = \frac{4}{12} = \frac{1}{3}$
- $P(\text{Blue} | B) = \frac{7}{15}$
- $P(\text{Blue} | C) = \frac{2}{15}$

Using Bayes' Theorem:

$$P(C | \text{Blue}) = \frac{P(\text{Blue} | C)P(C)}{P(\text{Blue})}$$
$$P(\text{Blue}) = \frac{1}{3} \cdot \frac{1}{4} + \frac{7}{15} \cdot \frac{1}{2} + \frac{2}{15} \cdot \frac{1}{4} = \frac{7}{20}$$
$$P(C | \text{Blue}) = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{7}{20}} = \frac{2}{21} \approx 0.0952$$

Answer: $\frac{2}{21} \approx 9.52\%$

2. Bayes Theorem (Section C)

Given:

- $P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$
- $P(\text{Blue} | A) = \frac{3}{10}$
- $P(\text{Blue} | B) = \frac{6}{15} = \frac{2}{5}$
- $P(\text{Blue} | C) = \frac{2}{15}$

$$P(\text{Blue}) = \frac{3}{10} \cdot \frac{1}{4} + \frac{2}{5} \cdot \frac{1}{2} + \frac{2}{15} \cdot \frac{1}{4} = \frac{37}{120}$$
$$P(C | \text{Blue}) = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{37}{120}} = \frac{4}{37}$$

Answer: $\frac{4}{37} \approx 10.81\%$

3. Medical Classification

Each class = 1800

Test data = 20%

Testing samples/class = 360

TP = 210

TN = 190

FP = 30

FN = 360 - 210 = 150

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN} = \frac{210 + 190}{580} = 68.97\%$$

Precision

$$= \frac{210}{210 + 30} = 87.5\%$$

Recall

$$= \frac{210}{210 + 150} = 58.33\%$$

Specificity

$$= \frac{190}{190 + 30} = 86.36\%$$

Answer

- FN = 150
- Accuracy = 68.97%
- Precision = 87.5%
- Recall = 58.33%
- Specificity = 86.36%

Effectiveness: High precision and specificity, but moderate recall means some cancer cases are missed.

4. Loss Analysis

i) Overfitting starts at Epoch 4

- ii) Early Stopping
- Dropout (or L2 Regularization)

iii) Overfitting reduces the model's performance on unseen data because it memorizes the training data instead of learning general patterns.

5. Accuracy Analysis

i)

Overfitting starts at Epoch 5

ii)

Training accuracy increases because the model memorizes training data, while validation accuracy decreases because it cannot generalize to new data.

iii)

- Early Stopping
- Data Augmentation (or Dropout)

6. Loss Analysis

i)

Best generalization: Epoch 4

ii)

Signs of overfitting:

- Training loss keeps decreasing.
- Validation loss starts increasing after Epoch 4.

iii)

- Dropout: Randomly removes neurons to prevent memorization.
- L2 Regularization: Reduces overly large weights and improves generalization.

7. Tire Classification

TP=119

FN=6

TN=120

FP=5

Accuracy

$$= \frac{239}{250} = 95.6\%$$

Precision

$$= \frac{119}{124} = 95.97\%$$

Recall

$$= \frac{119}{125} = 95.2\%$$

Specificity

$$= \frac{120}{125} = 96.0\%$$

F1

$$= 95.58\%$$

Answer

- Accuracy = 95.6%
- Precision = 95.97%
- Recall = 95.2%
- Specificity = 96.0%
- F1-score = 95.58%

8. Medical Diagnosis

TP=138

FN=12

TN=135

FP=15

Accuracy = 91%

Precision = 90.20%

Recall = 92%

Specificity = 90%

F1-score = 91.09%

9. Spam Detection

TP=180

FN=20

TN=170

FP=30

Accuracy = 87.5%

Precision = 85.71%

Recall = 90%

Specificity = 85%

F1-score = 87.80%

10. Manufacturing

TP=230

FN=20

TN=225

FP=25

Accuracy = 91%

Precision = 90.20%

Recall = 92%

Specificity = 90%

F1-score = 91.09%

11. Bayes Theorem (Shops)

Shop X:

$$P(A) = \frac{1}{3}$$

Shop Y:

$$P(B) = \frac{2}{3}$$

Adulterated probabilities:

$X = 30/50 = 0.6$

$Y = 20/60 = 1/3$

$$P(Y | Adulterated) = \frac{\frac{1}{3} \times \frac{2}{3}}{\frac{3}{5} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3}} = \frac{10}{19}$$

Answer: $\boxed{\frac{10}{19} \approx 52.63\%}$

12. Warehouse

Section A:

$$P(A) = \frac{3}{4}$$

Section B:

$$P(B) = \frac{1}{4}$$

Defective probabilities:

$$A = 15/50 = 0.3$$

$$B = 25/50 = 0.5$$

$$P(B \mid \text{Defective}) = \frac{0.5 \times 0.25}{0.3 \times 0.75 + 0.5 \times 0.25} = \frac{5}{14}$$

Answer: $\frac{5}{14} \approx 35.71\%$

13. Classification Metrics

Each class = 1000

20% test = 200/class

TP=120

TN=150

FP=50

FN=80

Accuracy = 67.5%

Precision = 70.59%

Recall = 60%

Specificity = 75%

14. Medical Classification

Each class = 1200

25% test = 300/class

TP=180

TN=160

FP=140

FN=120

Accuracy = 56.67%

Precision = 56.25%

Recall = 60%

Specificity = 53.33%

15. Loss Analysis

a) Best generalization: Epoch 4

b) Overfitting begins at Epoch 5 because validation loss starts increasing while training loss continues decreasing.

c)

- Early Stopping: Stops training when validation loss increases.
- Dropout: Randomly drops neurons to reduce overfitting.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand